

QUESTION BANK

SUBJECT :24MCADFCT102- DIGITAL FUNDAMENTALS & COMPUTER ARCHITECTURE

SEM/YEAR: I/I

Subject Handled By: Sumi M & Aparna A

Module I

Representation of signed numbers – 1's complement and 2's complement, Logic gates – AND - OR – NOT - NAND- NOR - XOR , Boolean algebra - Basic laws and theorems , Boolean functions - truth table, Standard forms of Boolean Expressions – Sum of Products and Product of Sums - minimization of Boolean function using Karnaugh map method Combinational Circuits - Half adder - Full Adder- Decoder -Encoder- Multiplexer – Demultiplexer

Q No	Question
1.	Differentiate the working of Encoders and Decoders by using proper example.
2.	Convert the given boolean expressions into standard POS form. (a) $F(A,B,C) = (A+B).(B+C).(A+C)$ (b) $F(A,B,C,D) = (A+C+D).(A+B+D).(A+C+D)$
3.	Minimize the following SOP expression using K-map (a) $BCD + ABCD + ABCD + ABCD + ABCD + ABCD + ABCD + ABCD + ABCD$ (b) $F(P,Q,R) = \sum_m(0,1,3,5,7)$
4.	What are logic gates? Explain different logic gates with their truth tables and diagram.
5.	Define the rules in Boolean algebra.
6.	Using Boolean algebra solve the given expression $AB+A(B+C)+B(B+C)$
7.	How can we represent the Boolean expressions in standard form?
8.	Convert the following SOP and POS expressions into truth table. (a) $ABC + ABC + ABC$ (b) $(A+B+C)(A+B+C)(A+B+C)$ (c) $ABC + ABC$ (d) $(A+B+C)(A+B+C)(A+B+C)(A+B+C)$
9.	Convert the given Boolean expressions into standard SOP form. (a) $ABC + AB + ABCD$ (b) $ABC + A B + ABCD$
10.	Explain the working of Multiplexer and De-multiplexer in detail
11.	A digital system receives an 8-bit 2's complement number 11010110 a) Determine its decimal value b) Convert it to 1's complement form
12.	Define half adder and full adder by including logical diagram and truth table.
13.	Explain basic laws and rules of Boolean algebra. Also simplify the given Boolean expression $AB + A(B+C) + B(B+C)$
14.	Represent the following numbers in Sign & Magnitude form, 1's compliment and 2's compliment form. (a) -75 (b) -125 (c) -98 (d) -119
15.	Simplify the following Boolean function in SOP form to obtain the minimal SOP expression.

	(a) $F(A,B,C,D) = \sum_m (0,1,3,5,7,6,10,13,14,15)$ (b) $F(A,B,C) = \sum_m (1,2,4,5,7)$
16.	Explain how signed numbers are represented using 1's and 2's complement form. Represent -15 in both 1's and 2's complement form
17.	Minimize the Boolean expression using k-map $F(W,X,Y,Z) = m(0,1,2,5,7,8,9,10,14)$
18.	Construct a de-multiplexer with four registers(R0-R3) and 2 serial inputs (S0-S1)

Module II

Sequential circuit - Clocking, Flip flops - SR – JK- D -T flip flops, Counters - Synchronous and asynchronous counters - UP/DOWN counters , Registers - Serial in serial out - Serial in parallel out - Parallel in serial out - Parallel in parallel out registers

Q No	Question
1.	Explain briefly about the working of 2-bit asynchronous up counter by using logic diagram and timing diagram
2.	What are the steps involved for designing a Synchronous counter? Design a 3 bit synchronous up counter using T-flipflop.
3.	Explain the working of D- flipflop and T- flipflop also spot the differences.
4.	What is sequential circuit? Give example.
5.	Show that a JK flip-flop can be converted to a D flip flop with an inverter between J and K.
6.	What is excitation table? List the excitation table for SR flip flop, JK flip flop, D flip flop and T flip flop.
7.	Differentiate SR and JK flipflops with neat diagrams.
8.	Explain Serial In Serial Out and Serial In Parallel Out Shift registers by using proper examples
9.	Define about Asynchronous counters. Design a 4 bit asynchronous down counter by using proper flipflop
10.	Explain about Shift registers and its major categories.
11.	Explain JK and D flipflops.
12.	Define asynchronous and synchronous counters.
13.	Differentiate Synchronous and Asynchronous Counters. Design a 4 bit Asynchronous Up counter with the help of JK flipflop
14.	Explain the application of shift registers and also differentiate Parallel in Serial Out and Parallel in Parallel Out shift registers
15.	Write a short note on a. Serial in parallel out registers b. Parallel in parallel out registers
16.	Design a 3-bit asynchronous up counter using T flipflop.

Module III

Introduction, Computer architecture -8 Design features, Application program - layers of abstraction, Five key components of a computer, Performance, Instruction set principles – Introduction, Classifying instruction set architectures, Memory addressing, Encoding an instruction set.

Q No	Question
1.	Explain the Architecture of Computer System
2.	Define Layers of Abstraction in Computer System
3.	List out and explain 8 Great Ideas in Computer Architecture
4.	Describe Five Key Components of a Computer
5.	Explain Instruction Codes
6.	Describe three Instruction Code Formats
7.	For a newly designed 32-bit processor, how should a CPU designer combine immediate, direct, indirect, and indexed addressing modes to achieve efficient array processing, constant handling, and procedure calls
8.	Define Reduced Instruction Set Computer
9.	Define Addressing Modes and different types of addressing modes
10.	Write a note on Instruction Cycle
11.	How to Encode an Instruction Set
12.	Write a short note on • Direct Addressing Mode • Indirect Addressing Mode
13.	List out and define the types of ISA
14.	Explain how a computer manufacturer can improve processor performance through CPU optimization and reduction in instruction count, without raising the clock frequency.
15.	Given two ISAs with equivalent capabilities but different instruction length variability, discuss why a compiler might choose one ISA rather than the other
16.	Define ISA in detail.

Module IV

The Processor - Introduction, Logic design conventions, Building a data path, A simple implementation scheme, An overview of pipelining - Pipelined data path and control - Structural hazards - Data hazards - Control hazards, interrupts – Input/output Interface-Mode of Transfer

Q No	Question
1.	In branch-intensive workloads, control hazards can significantly degrade performance. Explain this effect and examine how branch delay slots, branch prediction, and speculative execution address these penalties
2.	A processor produces wrong outputs when two successive instructions use the same register. What is the problem, and how can it be resolved
3.	In systems where DMA and CPU compete for memory access, recommend a strategy to avoid CPU performance degradation during large DMA operations.
4.	Explain • Logic Design Conventions • Types Of Logic Elements
5.	Discuss on Clocking Methodology
6.	Describe Data Path
7.	Explain Types of Elements In The Data path
8.	What is Pipelining
9.	Define 5 Clock Cycles in Pipelining
10.	Write a Short Note On • Data Hazard • Structural Hazard
11.	Define Modes of Transfers In Input/output Interface
12.	Explain DMA with neat diagram
13.	explain • Types of DMA Transfer • List out the advantages and Disadvantages of DMA
14.	Define Branch Hazard
15.	Define Pipelining. List out and explain the hazards in pipelining
16.	Differentiate Programmed I/O and Interrupt- initiated I/O

Module V

The Memory System – basic concepts, semiconductor RAM memories organization – static and dynamic RAM, Structure of larger memories, semiconductor ROM memories, Speed, Size and cost ,Cache memory – mapping functions – replacement algorithms

Q No	Question
1.	Explain Memory System
2.	Explain CPU-main memory connection with neat diagram
3.	What are the Techniques to Increase the Effective Size and Speed of the Memory
4.	Differentiate Cache Memory And Virtual Memory
5.	Explain Internal Organization of 16*8 Memory Chip With Neat Diagram
6.	Define SRAM Cell
7.	Differentiate Asynchronous Drams And Synchronous Drams with neat Diagram
8.	Explain Fast Page Mode
9.	Explain • Static Memories • Dynamic Memories
10.	Explain Memory Controller
11.	Explain Different Types of ROM
12.	Differentiate EPROM & EEPROM
13.	Explain Memory Hierarchy and define Characteristics of Memory Hierarchy
14.	Explain Cache Memory and define what are the characteristics of cache Memory
15.	Write a short note on • Cache Hit • Cache Miss • Read Hit • Write Hit • Read Miss • Write-Miss
16.	Explain the Application of Cache Memory
17.	What are the Advantages Disadvantages of Cache Memory
18.	Different Types of Cache Mapping Techniques
19.	Explain • Write-Through • Write-Back • Prefetching
20.	Describe Page Replacement Algorithms
21.	Given a 16 KB cache organized as a 2-way set associative cache with 32-byte blocks, design the cache structure and calculate the number of sets, index bits, and tag bits for a 32-bit address.
22.	A cache memory of 8 KB uses 2-way set associative mapping with 16-byte blocks. Determine the number of sets, tag bits, and index bits for a 32-bit address.”
23.	Propose a memory hierarchy for a modern computing system. Discuss the role of each memory level with respect to speed, cost, and storage capacity, and explain how cache hit ratio and memory access latency affect performance.